Best-of-N or Tree Search? A Controlled Study of Learned-Dynamics Optimism Under Test-Time Planning

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Abstract

Tree search over a learned dynamics model is usually framed as planning. This paper studies a complementary view: adaptive search is also a test-time optimizer over the model, and can therefore concentrate compute on model errors. We introduce a controlled continuous-control point-mass task with a localized optimistic bias pocket in a dynamics and reward surrogate. Static best-of-N evaluates independent rollouts, while UCT MCTS adaptively expands and backs up branches that look valuable under the surrogate. In a repeated-seed full run, UCT MCTS exhibits a larger selected-return optimism gap than best-of-N at the largest tested forward-model budget, and uncertainty-penalized search reduces the gap below both. The contribution is not a robotics benchmark result; it is a compact mechanism testbed for studying when adaptive planning improves behavior and when it amplifies model optimism.

1 Introduction

Learned model-based agents such as PlaNet, Dreamer, and PETS show that learned dynamics can support efficient control (Hafner et al., 2019; 2020; Chua et al., 2018). Monte Carlo tree search (MCTS) variants are also being extended to imperfect or continuous transition models (Kohankhaki et al., 2024; Riviere et al., 2024). These systems usually treat search as a way to spend extra computation at test time in order to improve action selection.

This paper studies a failure mode of that framing. If the dynamics or reward model is also the test-time evaluator, then adaptive planning is optimizing a fallible objective. More model queries can discover better plans, but they can also discover and repeatedly exploit localized optimism. We isolate this mechanism in a small deterministic control problem where the true environment and the biased surrogate are both known, making the optimism gap directly measurable.

The central question is narrow: under equal forward-model budgets, can adaptive tree search select plans whose model-predicted return is more optimistic than plans selected by static best-of-N sampling? In the controlled setting studied here, the answer is yes for UCT MCTS at the largest tested budget. We also test two uncertainty-based repairs that lower the selected-return gap in the same artifact.

2 Related Work

Model-based reinforcement learning has long used learned transition models for planning and policy improvement. PETS uses probabilistic ensembles with model-predictive control (Chua et al., 2018), while PlaNet and Dreamer learn latent dynamics that support planning or policy learning by imagination (Hafner et al., 2019; 2020). These successes motivate using learned dynamics as a source of test-time computation, but they also expose agents to model bias when planning aggressively.

MCTS is a canonical adaptive search procedure, and recent work studies tree search under transition uncertainty and continuous dynamics (Kohankhaki et al., 2024; Riviere et al., 2024). Separately, test-time compute work distinguishes independent best-of-N sampling from adaptive search against

Table 1: Mean diagnostics at budget 1024. Lower optimism gap is better. Root concentration is the fraction of root visits assigned to the most visited child and is defined only for tree-search methods.

Planner	Optimism gap	Root concentration
Best-of-N	0.301	–
UCT MCTS	0.686	0.649
Value MCTS	0.397	0.316
Uncertainty MCTS	0.236	0.727
Conservative MCTS	0.261	0.729

a learned scorer (Snell et al., 2024). Our experiment connects these views by asking how adaptive allocation changes exposure to model optimism when the evaluator is imperfect.

3 Method

3.1 Controlled Environment

The environment is a deterministic two-dimensional point robot. The true reward encourages progress to a goal, penalizes action cost, and mildly penalizes an off-route pocket. The surrogate dynamics and reward model is accurate almost everywhere but, near that pocket, predicts extra reward and a transition drift toward the goal. It also reports elevated uncertainty in the same region. This makes the failure mode visible without claiming that the surrogate was learned from data.

For a selected action sequence $a_{0:H-1}$, we measure

$$\Delta_{\text{opt}} = \hat{G}(a_{0:H-1}) - G(a_{0:H-1}),$$

where $\hat{G}$ is the surrogate rollout return and G is the true rollout return for the same sequence. Lower Δ_{opt} is better. We also record cumulative reward bias, transition error, uncertainty exposure, root visit concentration, and depth-wise reward bias.

3.2 Planners

All planners use the same discrete action set, horizon $H = 18$, discount $\gamma = 0.98$, and forward-model budget accounting. The suite includes a random open-loop baseline, static best-of-N rollout selection, UCT MCTS, value-guided MCTS, uncertainty-penalized MCTS, and conservative-backup MCTS. Best-of-N samples independent open-loop trajectories and selects the maximum surrogate return. MCTS adaptively expands and backs up branches according to UCB-style selection.

The intended repair is a lower-confidence search target. If the uncertainty signal $u(s, a)$ tracks local optimism, replacing $\hat{r}$ with $\hat{r} - \lambda u$ during scoring or backup can suppress the optimistic branch before it dominates root statistics. This is a controlled calibration assumption rather than a general statistical guarantee.

4 Experiments

The full run uses 20 random seeds and five budget settings from 64 to 1024. The committed artifacts include per-plan metrics, closed-loop episode diagnostics at the largest budget, root statistics, and the figures used below. The headline result is recorded in `results/full/manifest.json`, with mean diagnostics in `results/full/summary.csv`.

Table 1 shows the largest-budget slice. UCT MCTS has a selected-return optimism gap of 0.686, compared with 0.301 for static best-of-N. The strongest uncertainty-penalized repair in this run reduces the gap to 0.236, below both UCT MCTS and best-of-N.

Figure 1 plots the optimism gap over budgets. Figure 2 shows that tree search increasingly concentrates visits at the root as budgets grow. Figure 3 gives a depth-wise view of reward bias along

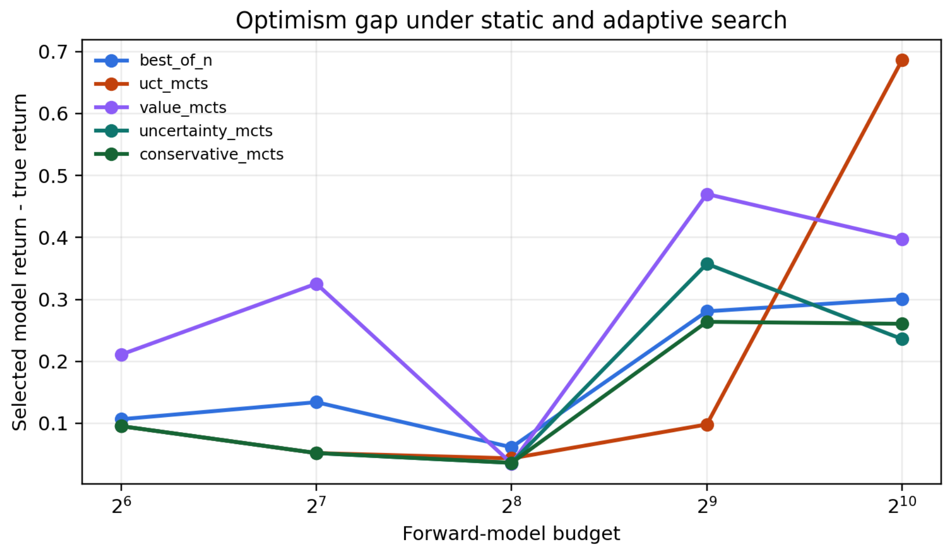

Figure 1: Selected-return optimism gap across forward-model budgets.

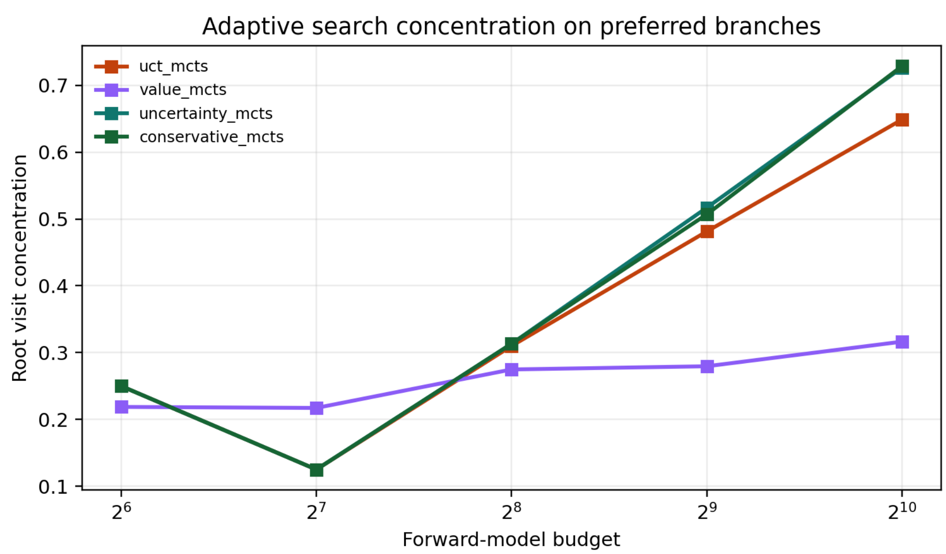

Figure 2: Root visit concentration as adaptive tree search allocates budget to preferred branches.

selected plans. Together, these diagnostics support the mechanism claim that adaptive search can allocate additional computation toward model-optimistic branches.

5 Reproducibility

The accompanying artifact is organized around relative paths only. The full experiment is produced by `experiments/run_mechanism.py` with mode `full` and output `results/full`; the smoke run changes the mode and output path to `smoke` and `results/smoke`. The experiments use deterministic seeds $0, \ldots, 19$ for the full run, and the generated CSV/JSON/PNG outputs

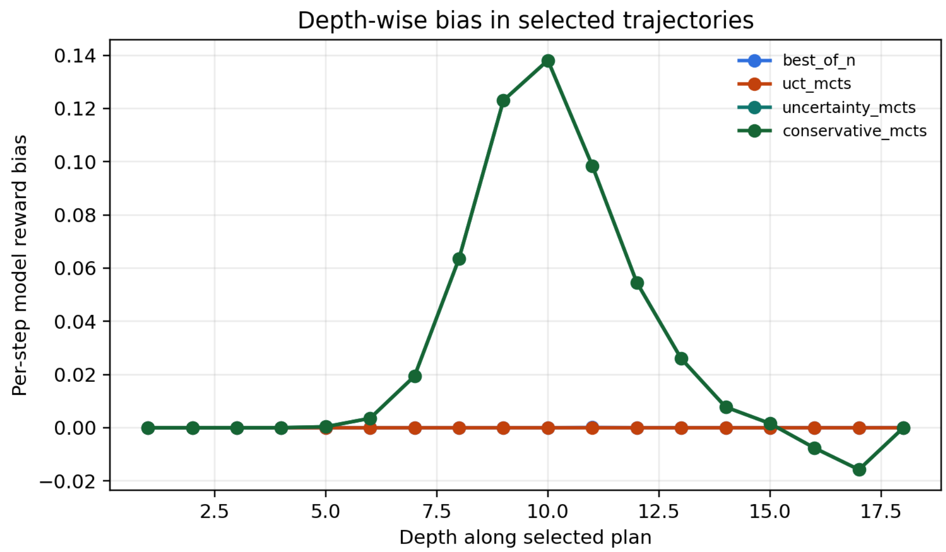

Figure 3: Depth-wise surrogate reward bias along selected plans in the largest-budget run.

are committed under `results/full/` and `paper/figures/`. The paper build is scripted by `scripts/build_paper.ps1`.

6 Discussion

The result supports a narrow mechanism claim: in this controlled model, adaptive UCT search can select trajectories whose surrogate return is more optimistic than trajectories selected by static best-of-N under equal model-step budgets. The uncertainty-penalized variants reduce the selected-return gap in the largest-budget run, suggesting that uncertainty-aware objectives can be useful when uncertainty is aligned with model optimism.

The experiment should be read as a microscope, not a benchmark. Its value is that the failure mode is observable, testable, and easy to extend. It does not establish that the same effect dominates in high-dimensional robotics tasks, nor that the uncertainty penalty transfers across model classes.

7 Limitations and Next Steps

The surrogate model is hand-designed rather than trained from data. The uncertainty signal is intentionally aligned with the injected bias pocket. The curves are seed-sensitive because the task is deliberately small, and no statistical claim beyond the committed repeated-seed run is made here.

The next version should train an ensemble dynamics model on biased or off-policy data, add continuous-action expansion or sampling-based proposals, and evaluate on at least one standard continuous-control benchmark. The repair should be stress-tested under miscalibrated uncertainty and value-function optimism. The most important open question is whether search concentration predicts real downstream regret when the model is learned rather than specified.

References

Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, 2018. URL https://arxiv.org/abs/1805.12114.

Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. *arXiv preprint arXiv:1811.04551*, 2019. URL https://arxiv.org/abs/1811.04551.

Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. *arXiv preprint arXiv:1912.01603*, 2020. URL https://arxiv.org/abs/1912.01603.

Farnaz Kohankhaki, Kiarash Aghakasiri, Hongming Zhang, Ting-Han Wei, Chao Gao, and Martin Müller. Monte carlo tree search in the presence of transition uncertainty. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 20151–20158, 2024. doi: 10.1609/aaai.v38i18.29994. URL https://ojs.aaai.org/index.php/AAAI/article/view/29994.

Benjamin Riviere, John Lathrop, and Soon-Jo Chung. Monte carlo tree search with spectral expansion for planning with dynamical systems. *Science Robotics*, 2024. URL https://arxiv.org/abs/2412.11270.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024. URL https://arxiv.org/abs/2408.03314.